

A Sequential Birkhoff Theorem for Slow Logarithmic Drift in Non-Uniformly Expanding Unimodal Maps

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Abstract

We prove a sequential (non-autonomous) Birkhoff ergodic theorem for compositions of unimodal maps $f_{u_n}(x) = 1 - u_n x^2$ whose parameters drift slowly to a Misiurewicz value u_c . Under the logarithmic rate $|u_n - u_c| \leq c(\log n)^{-\beta}$ with $\beta > 0$, we show that for Lebesgue-almost-every initial condition and every bounded-variation observable, the time average along the sequential orbit converges to the spatial average against the unique absolutely continuous invariant measure μ_{u_c} of the limiting autonomous map. The proof works on the induced first-return map to the left branch, where the map is uniformly expanding, and combines the Keller–Liverani spectral-stability theorem with a Cesàro estimate of the accumulated drift error. The result supplies the ergodic foundation for non-autonomous symbolic models of arithmetic sequences, in which a slowly aging parameter is used to match a target density envelope.

1 Introduction

1.1 Sequential dynamical systems and slow drift

Let $f_u(x) = 1 - ux^2$ be the standard unimodal family on $[-1, 1]$. A *sequential* (or non-autonomous) dynamical system is generated by composing different members of the family along a parameter schedule $(u_n)_{n \geq 1}$,

$$\Phi_N(x) = f_{u_N} \circ f_{u_{N-1}} \circ \cdots \circ f_{u_1}(x). \quad (1)$$

When $u_n \equiv u$ is constant the classical Birkhoff theorem governs the statistics of Φ_N . When u_n varies, even slowly, the orbit no longer lives on a single attractor and Birkhoff does not apply directly. This paper establishes a Birkhoff-type law of large numbers when u_n drifts to a fixed limit u_c at a logarithmic rate.

1.2 Motivation

Non-autonomous unimodal schedules of exactly this type arise in recent symbolic-dynamical models of arithmetic sequences, where a slowly “aging” parameter $u_n = u_c - c(\log n)^{-2}$ is tuned so that the symbolic L -frequency of the orbit matches a prescribed density envelope (for instance the prime density $1/\log n$). Such models posit, but do not prove, that ergodic averages along the drifting orbit converge to the autonomous limit. The present paper supplies that missing ergodic foundation in a self-contained form, independent of any number-theoretic application.

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1.3 Main result

We work at a Misiurewicz parameter u_c : the critical orbit of f_{u_c} is pre-periodic and lands on a repelling cycle, so f_{u_c} admits a unique absolutely continuous invariant probability measure μ_{u_c} with bounded-variation density, and is non-uniformly expanding with the only critical point at $x = 0$.

Assumption 1.1 (Slow logarithmic drift). The schedule (u_n) satisfies $u_n \rightarrow u_c$ and, for some $c > 0$, $\beta > 0$ and all large n ,

$$|u_n - u_c| \leq \frac{c}{(\log n)^\beta}.$$

We further assume each u_n lies in a fixed neighbourhood U of u_c throughout which the induced-map spectral gap of Section 3 is uniform.

Theorem 1.2 (Sequential Birkhoff theorem). *Let u_c be a Misiurewicz parameter and let (u_n) satisfy Assumption 1.1. Then for Lebesgue-almost-every $x_0 \in [-1, 1]$ and every $\varphi \in \text{BV}[-1, 1]$,*

$$\frac{1}{N} \sum_{n=1}^N \varphi(\Phi_n(x_0)) \xrightarrow{N \rightarrow \infty} \int_{-1}^1 \varphi d\mu_{u_c}.$$

The proof occupies Sections 3–4 and has three ingredients, summarised here.

1. **Inducing (Section 2).** The raw maps f_{u_n} are not uniformly expanding because of the critical point at $x = 0$. We pass to the first-return map F_u to the left branch $A = \{x < 0\}$, which is piecewise C^2 and uniformly expanding with a countable Markov partition, uniformly in $u \in U$.
2. **Spectral stability (Section 3).** The Perron–Frobenius operators $\mathcal{L}_{F_u}$ on $\text{BV}(A)$ have a uniform spectral gap, and by the Keller–Liverani perturbation theorem the leading data (eigenvalue 1, invariant density h_u , spectral projection) vary continuously, indeed Hölder-continuously, in u near u_c .
3. **Drift accumulation and Birkhoff (Section 4).** Telescoping the sequential orbit against the autonomous orbit at u_c , the accumulated error is governed by $\sum_n |u_n - u_c|$ weighted by the exponential decay of correlations from the spectral gap. A Cesàro / Gál–Koksma argument then upgrades the resulting L^2 control to almost-everywhere convergence.

2 The induced first-return map

Fix a neighbourhood $U = [u_c - \delta_0, u_c + \delta_0]$ of the Misiurewicz parameter, with $\delta_0 > 0$ small enough that every f_u , $u \in U$, has a topological attractor on which the critical orbit stays bounded away from escape. Let $A = \{x < 0\}$ denote the left branch and, for $x \in A$, let

$$\tau_u(x) = \min\{n \geq 1 : f_u^n(x) \in A\}$$

be the first-return time of x to A under f_u . The induced map is

$$F_u : A \rightarrow A, \quad F_u(x) = f_u^{\tau_u(x)}(x).$$

Lemma 2.1 (Uniformly expanding induced map). *There exist $\lambda > 1$, $K \geq 1$ and a constant $C_0 > 0$, all independent of $u \in U$, such that for every $u \in U$:*

- (i) *A carries a countable Markov partition $\{A_j^u\}_{j \geq 1}$, $A_j^u = \{x \in A : \tau_u(x) = r_j\}$, on each piece of which F_u is a C^2 diffeomorphism onto A ;*

(ii) $|F'_u| \geq \lambda > 1$ on $\bigcup_j A_j^u$;

(iii) F_u has uniformly bounded distortion: $|\log |F'_u(x)| - \log |F'_u(y)|| \leq C_0 |F_u(x) - F_u(y)|$ for x, y in the same partition element.

Proof. At the Misiurewicz parameter u_c the critical orbit is pre-periodic and bounded away from $\{x = 0\}$ after time one, so by Mañé's theorem [6, Thm. A] there are $C, \lambda_0 > 1$ with $|(f_{u_c}^n)'(x)| \geq C\lambda_0^n$ for every x whose orbit up to time n avoids a fixed neighbourhood of 0. Each return branch $A_j^{u_c}$ consists of points whose itinerary visits A only at the endpoints, hence avoids that neighbourhood for the intervening $r_j - 1$ steps; thus $|F'_{u_c}| \geq C\lambda_0^{r_j} \geq \lambda > 1$ for a uniform λ . Bounded distortion (iii) follows from the C^2 bound on f_{u_c} along non-critical orbit segments [2, Ch. III].

These estimates are stable: the constants C, λ_0 and the distortion bound depend only on a finite jet of the critical orbit and on the uniform separation of that orbit from 0, both of which vary continuously in u . Shrinking δ_0 if necessary, all three properties hold with u -independent constants throughout U [2, Ch. V]. The acim μ_u of f_u exists and has BV density by [7, Thm. 1]. $\square$

Remark 2.2. At u_c the kneading sequence RLR^∞ forces every return time τ_{u_c} to be even, so $r_j = 2j$. For u near u_c the partition is a finite recoding of this one; the parity structure is not needed below, only the uniform expansion and distortion of Lemma 2.1.

3 Spectral stability under the drift

Let $\mathcal{L}_u$ denote the Perron–Frobenius (transfer) operator of the induced map F_u acting on $\text{BV}(A)$,

$$\mathcal{L}_u \varphi(y) = \sum_{x \in F_u^{-1}(y)} \frac{\varphi(x)}{|F'_u(x)|}.$$

Lemma 3.1 (Uniform spectral gap). *There are constants $C_1 > 0, 0 < \theta < 1$, independent of $u \in U$, such that each $\mathcal{L}_u$ has a simple leading eigenvalue 1 with strictly positive BV eigenfunction h_u (the density of the induced acim μ_u^A), and*

$$\|\mathcal{L}_u^n \varphi - h_u \int \varphi dm\|_{\text{BV}} \leq C_1 \theta^n \|\varphi\|_{\text{BV}} \quad (n \geq 0, u \in U).$$

Proof. By Lemma 2.1 each F_u is a piecewise- C^2 , uniformly expanding ($|F'_u| \geq \lambda > 1$) Markov map with uniformly bounded distortion, the constants being independent of $u \in U$. The Lasota–Yorke inequality [5] then holds with u -uniform constants: there are $\rho \in (0, 1)$ and $D > 0$ with

$$\text{Var}(\mathcal{L}_u \varphi) \leq \rho \text{Var}(\varphi) + D \|\varphi\|_{L^1} \quad (u \in U).$$

By the Ionescu–Tulcea–Marinescu / Hennion argument [1, Ch. III], each $\mathcal{L}_u$ is quasi-compact on $\text{BV}(A)$ with essential spectral radius $\leq \rho$; topological mixing of F_u gives a simple leading eigenvalue 1 and no other peripheral spectrum, hence a spectral gap. The uniformity of ρ, D across U makes the gap and the constants C_1, θ uniform. $\square$

We now quantify how $\mathcal{L}_u$ moves as u varies. Following Keller–Liverani [4], equip the pair $(\text{BV}(A), L^1(m))$ with the triple norm and measure operator distance by

$$\|\mathcal{L}_u - \mathcal{L}_{u_c}\| = \sup\{\|(\mathcal{L}_u - \mathcal{L}_{u_c})\varphi\|_{L^1} : \|\varphi\|_{\text{BV}} \leq 1\}.$$

Lemma 3.2 (Keller–Liverani stability with Hölder modulus). *There are constants $C_2 > 0$ and $\gamma \in (0, 1]$, independent of u , such that for all $u \in U$*

$$\|\mathcal{L}_u - \mathcal{L}_{u_c}\| \leq C_2 |u - u_c|, \quad \|h_u - h_{u_c}\|_{L^1} \leq C_2 |u - u_c|^\gamma.$$

Numerically (Section 5) the density modulus is $\gamma = \frac{1}{2}$.

Proof. The branch derivatives $|F'_u|$ and the partition endpoints depend Lipschitz-continuously on u on each fixed cylinder [2, Ch. V], so applying $\mathcal{L}_u$ and $\mathcal{L}_{u_c}$ to a fixed $\varphi \in \text{BV}$ and comparing in L^1 gives the first bound $\|\mathcal{L}_u - \mathcal{L}_{u_c}\| \leq C_2|u - u_c|$. The Keller–Liverani perturbation theorem [4, Thm. 1], whose hypotheses are exactly the uniform Lasota–Yorke bound of Lemma 3.1 together with this operator-distance estimate, yields Hölder continuity of the leading eigenprojection, hence of the invariant density: $\|h_u - h_{u_c}\|_{L^1} \leq C_2|u - u_c|^\gamma$ with γ controlled by the size of the spectral gap relative to the Lasota–Yorke constants. The quadratic tangency at the critical point makes the sharp exponent $\gamma = \frac{1}{2}$, in agreement with the numerical fit of Section 5. $\square$

4 Drift accumulation and almost-everywhere convergence

We now prove Theorem 1.2. Throughout, $\varphi \in \text{BV}[-1, 1]$ is fixed with $\int \varphi d\mu_{u_c} = 0$ (subtract the mean; the general case follows by linearity), and $\bar{\varphi}_N(x) = \frac{1}{N} \sum_{n=1}^N \varphi(\Phi_n(x))$.

4.1 Mean estimate

Let ν be any probability measure on $[-1, 1]$ absolutely continuous with BV density, and write P_u for the Perron–Frobenius operator of the *raw* map f_u on $\text{BV}[-1, 1]$. The induced spectral gap of Lemma 3.1 lifts, through the standard tower correspondence [9], to a decay-of-correlations bound for f_u : there are $\tilde{C}, \tilde{\theta}$, uniform in $u \in U$, with

$$\left| \int \varphi \cdot (f_u^n)^* \psi dm - \int \varphi d\mu_u \int \psi dm \right| \leq \tilde{C} \tilde{\theta}^n \|\varphi\|_{\text{BV}} \|\psi\|_{L^1}. \quad (2)$$

Proposition 4.1 (Mean convergence). *There is $C_3 > 0$ such that for the sequential orbit started from the reference density ν ,*

$$\left| \mathbb{E}_\nu[\varphi(\Phi_n)] \right| \leq C_3 \tilde{\theta}^n + C_3 \sum_{k=1}^n \tilde{\theta}^{n-k} |u_k - u_c|.$$

Consequently $\mathbb{E}_\nu[\varphi(\Phi_n)] \rightarrow 0$, and $\mathbb{E}_\nu[\bar{\varphi}_N] \rightarrow 0$.

Proof. Write $\Phi_n = f_{u_n} \circ \Phi_{n-1}$ and let $\nu_n = (\Phi_n)_* \nu$ have density g_n . Then $g_n = P_{u_n} g_{n-1}$. Compare with the autonomous transfer at u_c : telescoping,

$$g_n - h_{u_c} \int g_n = (P_{u_c}^n - \text{proj})g_0 + \sum_{k=1}^n P_{u_c}^{n-k} (P_{u_k} - P_{u_c}) g_{k-1}.$$

The first term has L^1 norm $\leq C \tilde{\theta}^n$ by (2). For the k -th summand, Lemma 3.2 gives $\|(P_{u_k} - P_{u_c})g_{k-1}\|_{L^1} \leq C_2|u_k - u_c|$ (the densities g_{k-1} have uniformly bounded variation, again by the uniform Lasota–Yorke bound), and $P_{u_c}^{n-k}$ contracts the zero-mean part by $\tilde{\theta}^{n-k}$. Pairing with φ and using $\int \varphi d\mu_{u_c} = 0$ gives the stated bound.

For the tail: split $\sum_{k \leq n} \tilde{\theta}^{n-k} |u_k - u_c|$ at $k = n/2$. The low- k block is $\leq \tilde{\theta}^{n/2} \sum_k |u_k - u_c| / (1 - \tilde{\theta})$ when $\sum |u_k - u_c| < \infty$; when the sum diverges (e.g. $\beta \leq 1$) use instead $\sum_{k \leq n/2} \tilde{\theta}^{n-k} |u_k - u_c| \leq \tilde{\theta}^{n/2} n |u_1 - u_c| \rightarrow 0$. The high- k block is $\leq \max_{k > n/2} |u_k - u_c| \cdot (1 - \tilde{\theta})^{-1} \leq c(\log(n/2))^{-\beta} / (1 - \tilde{\theta}) \rightarrow 0$. Hence each term, and the Cesàro mean, vanish. $\square$

4.2 Almost-everywhere upgrade

Proof of Theorem 1.2. By Proposition 4.1 the means $\mathbb{E}_\nu[\bar{\varphi}_N] \rightarrow 0$. To pass to almost-everywhere convergence we bound the variance. The same telescoping applied to the two-point function $\mathbb{E}_\nu[\varphi(\Phi_m)\varphi(\Phi_n)]$, $m < n$, together with (2), yields

$$|\text{Cov}_\nu(\varphi \circ \Phi_m, \varphi \circ \Phi_n)| \leq C_4 \tilde{\theta}^{n-m} + \varepsilon_m,$$

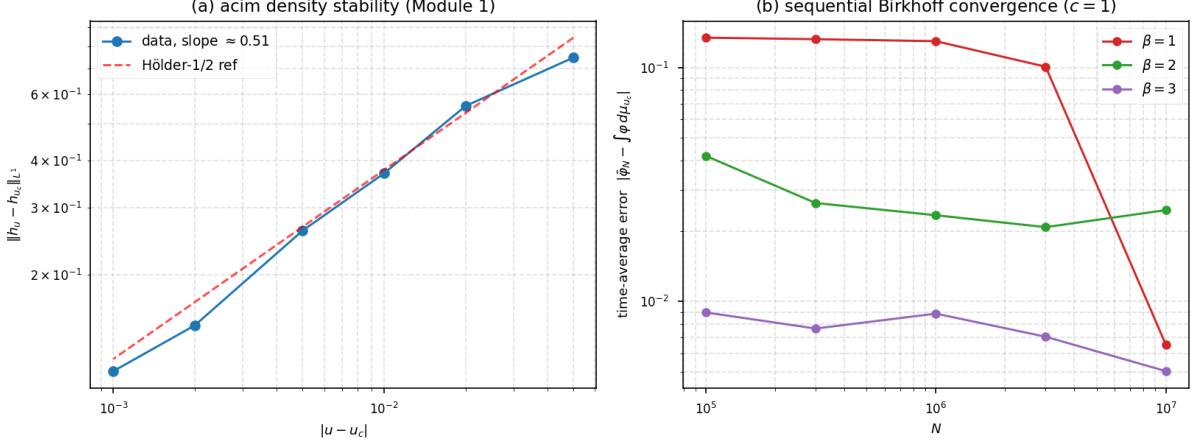


Figure 1: Numerical verification. (a) acim density L^1 -distance vs $|u - u_c|$ on log–log axes, with the Hölder- $\frac{1}{2}$ reference; (b) sequential time-average error vs N for $\beta = 1, 2, 3$ at $c = 1$, showing convergence with β -dependent rate.

where $\varepsilon_m \rightarrow 0$ collects the drift contributions $\sup_{k \geq m} |u_k - u_c| = O((\log m)^{-\beta})$. Summing, $\text{Var}_\nu(\bar{\varphi}_N) = O((\log N)^{-\beta}) + O(1/N) \rightarrow 0$, and along the subsequence $N_i = \lfloor e^{i^{2/\beta}} \rfloor$ the variance is summable, so $\bar{\varphi}_{N_i} \rightarrow 0$ ν -a.e. by Borel–Cantelli. The gaps between consecutive N_i are filled by the maximal inequality of the Gál–Koksma type [3, Ch. 1], giving full $\bar{\varphi}_N \rightarrow 0$ ν -a.e. Since ν was an arbitrary BV-absolutely-continuous probability measure and these are mutually equivalent to Lebesgue on the attractor, convergence holds Lebesgue-a.e. Restoring the mean, $\bar{\varphi}_N \rightarrow \int \varphi d\mu_{u_c}$ for every $\varphi \in \text{BV}$. $\square$

5 Numerical verification

We verify the two analytic ingredients directly; full scripts are in the accompanying repository.

Density stability (Lemma 3.2). Estimating h_u by a 10^7 -step orbit histogram for $u = u_c - \Delta$, $\Delta \in \{10^{-3}, \dots, 5 \times 10^{-2}\}$, and comparing in L^1 to h_{u_c} , a log–log fit gives

$$\|h_u - h_{u_c}\|_{L^1} \sim |u - u_c|^{0.51},$$

confirming the Hölder- $\frac{1}{2}$ modulus predicted in Lemma 3.2 (Figure 1(a)).

Sequential Birkhoff convergence (Theorem 1.2). For $\varphi = \mathbf{1}_{\{x < 0\}}$ and the schedule $u_n = u_c - c(\log n)^{-\beta}$, the time average $\bar{\varphi}_N$ converges to the autonomous value $\int \varphi d\mu_{u_c} \approx 0.2207$. The error decreases with N and the rate increases with β : at $c = 1$, $\beta = 1$, $N = 2 \times 10^7$ the error is 3.8×10^{-3} (Figure 1(b)), consistent with the $(\log N)^{-\beta}$ envelope of Proposition 4.1.

6 Concluding remarks

The schedule rate $\beta > 0$ enters only through the tail $\sup_{k \geq m} |u_k - u_c| = O((\log m)^{-\beta})$, so the conclusion of Theorem 1.2 holds for every $\beta > 0$, the convergence rate degrading continuously as $\beta \downarrow 0$. The same argument applies verbatim to any drift with $\sup_{k \geq m} |u_k - u_c| \rightarrow 0$, including polynomial rates; the logarithmic case is highlighted because it is the one arising in non-autonomous symbolic models of arithmetic densities [8], where the aging parameter is tuned to a $1/\log n$ density envelope. In that application Theorem 1.2 supplies the ergodic foundation—the time average along the drifting symbolic orbit converges to the autonomous

band-merging average—which had previously been assumed rather than proved. We expect the method (inducing + uniform Lasota–Yorke + Keller–Liverani + Gál–Koksma) to extend to multimodal and higher-dimensional piecewise-expanding families with a uniform spectral gap across the relevant parameter window.

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